INCH-POUND
MIL-C-48487C(AR)
09 May 1995
(see 6.4)

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY - LAMP DIODES

This specification is approved for use by the U.S. Army Armament, Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly Lamp Diodes, 11732539 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use to improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

MIL-I-45607 - Inspection Equipment; Acquisition, Maintenance, and Disposition of

STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods
MIL-STD-2000 - Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-2073-1 - Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on date of the solicitation.

U.S. Army Armament Research Development and Engineering Center (ARDEC)

DRAWINGS

11732539 - Circuit Card Assembly - Lamp Diodes

(Copies of Government drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained (see control provision for additional precedence criteria).

3. REQUIREMENTS

3.1 Fabrication. The lamp diodes circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732539 and drawings pertaining thereto

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

and, when assembled, shall meet the requirements of this specification (see 4.5.1).

3.1.1 Function. The assembly shall provide the following functions:

a. Provide diode logic for testing the Field Test Set card fail lamps.

b. Contain scaling resistors for testing Gunner's Control Unit (GCU) potentiometer deflections.

c. Contain bridge resistance for testing GCU laser analog range scaling resistors.

3.1.2 General specifications. (see 4.6.3)

3.1.2.1 Manufacture and inspection. The following provisions of MIL-F-13926 apply:

a. Dimensions and tolerances
b. Effect of protective coating on dimensions (Inorganic type coatings)
c. Part identification and marking
d. Workmanship (including applicable portions of MIL-STD-2000 and MIL-STD-454, Requirement 9)

3.1.2.2 Standards of manufacture.

3.1.2.2.1 Assembly and soldering. The requirements of MIL-STD-2000 shall apply, as a minimum.

3.1.2.2.2 Interchangeability. The assembly shall be manufactured in accordance with interchangeability requirements as specified in MIL-STD-100.

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° ± 18°F
b.	Relative humidity	50 percent ±30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified in the contract or purchase order (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.3.1 Lamp diodes forward biased. With the interconnection (item 3.1 of Table I) applied, connect the return of the power supply (item 2.1 of Table I) to pin P1-A20 and the output of the power supply to one side of the load lamp (item 1.1 of Table I). Connect the other side of the load lamp to each of the pins specified by Table II, indvidually, in any item number order. The load lamp shall illuminate for each connection (see 4.6.1.1).

3.3.2 Lamp diodes reverse biased. With the interconnection (item 3.1 of Table I) applied, connect the output of the power supply (item 2.1 of Table I) to one side of the load lamp (item 1.1 of Table I) and connect the other side of the load lamp to pin P1-A20. Connect the return of the power supply to each of the pins specified by Table II, individually, in any item number order. The load lamp shall remain extinguished for each connection (see 4.6.1.2).

TABLE I. Loads, power and interconnection.

Item	Conditions	Characteristics	Connections
1.0	Load		
1.1	Lamp	+24V dc lamp	As specified
2.0	Power source		
2.1	24V dc	Tolerance: ±2V dc Ripple: 200mV p-p max Current: 500ma min	As specified
2.2	5.25V dc	Tolerance: ±0.15V dc Ripple: 20mV p-p max Current: 500ma min	When specified Applied between Pins P1-B13(+) and P1-B17(-)
3.0	Interconnections		Interconnections of listed pins:
3.1	-	-	P1-A20 to P1-B20
3.2	-	-	P1-B13 to P1-B17

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the currentversion before use.

TABLE II. Lamp diodes verification.

Item	Connection of 24V Lamp to P1 Pin	Item	Connection of 24V Lamp to P1 Pin
1	B11	15	A21
2	A12	16	B21
3	B12	17	B22
4	A14	18	A24
5	B14	19	A25
6	A15	20	B25
7	B15	21	A26
8	A16	22	B26
9	B16	23	A27
10	A17	24	B27
11	A18	25	A28
12	B18	26	B28
13	A19	27	A29
14	B19	28	B29

3.3.3 Scaling and bridge resistors. The resistance between pins as listed in Table III shall be as specified therein (see 4.6.1.3).

TABLE III. Scaling and bridge resistance verification.

Item	Between P1 Pins	Resistance
1	B1 and A7	150 ± 1.5 kilohms
2	A2 and B7	150 ± 1.5 kilohms
3	B2 and A8	69.8 kilohms ± 698 ohms
4	B3 and B8	150 ± 1.5 kilohms
5	B4 and A9	10.0 ± 0.100 kilohms
6	A5 and B9	10.0 ± 0.100 kilohms
7	B5 and A10	10.0 ± 0.100 kilohms
8	A6 and A11	10.0 ± 0.100 kilohms
9	B6 and B10	100 ± 1 kilohms
10	A13 and B23	100 ± 1 kilohms
11	A4 and B23	750 ± 7.5 ohms
12	A3 and A23	100 ± 1 kilohms
13	A1 and A23	150 ± 1.5 kilohms
14	A22 and B24	3970 ± 1.985 ohms

3.3.4 Laser range finder select logic. With the power source (item 2.1 of Table I) applied between pins P1-A30 (+) and P1-B17 (-) and: (see 4.6.1.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

a. With the power source (item 2.2 of Table I) applied, the output at pin P1-B30 shall be 0.0 ± 0.5V dc measured with respect to P1-B17.

b. With the interconnection (item 3.2 of Table I) applied the output at P1-B30 shall be +24 ± 2V dc measured with respect to P1-B17.

3.4 Environmental.

3.4.1 Temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of +40°F to 100°F and shall not be damaged when subjected to the storage conditions ranging from -65°F to 160°F (see 4.6.2.1 and 4.6.2.2).

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to a sinusoidal vibration displacement of .03 inch double amplitude, 10 - 55 Hz in accordance with Figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes (see 4.6.2.3).

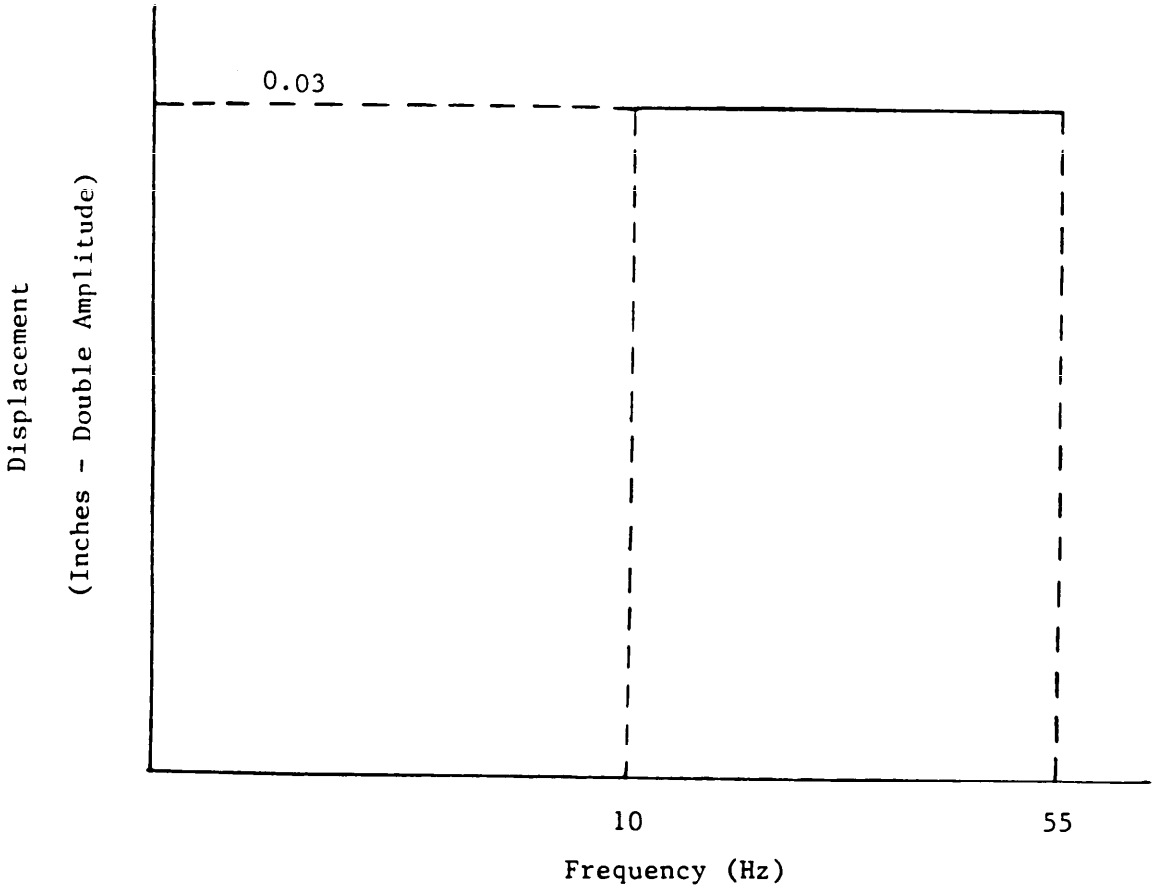


FIGURE 1. Vibration Curve.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the currentversion before use.

3.4.3 Shock. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) in accordance with Figure 2. The assembly shall withstand three shock pulses applied in both directions along three mutually perpendicular axes (18 shocks). In addition the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g's for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes (see 4.6.2.4).

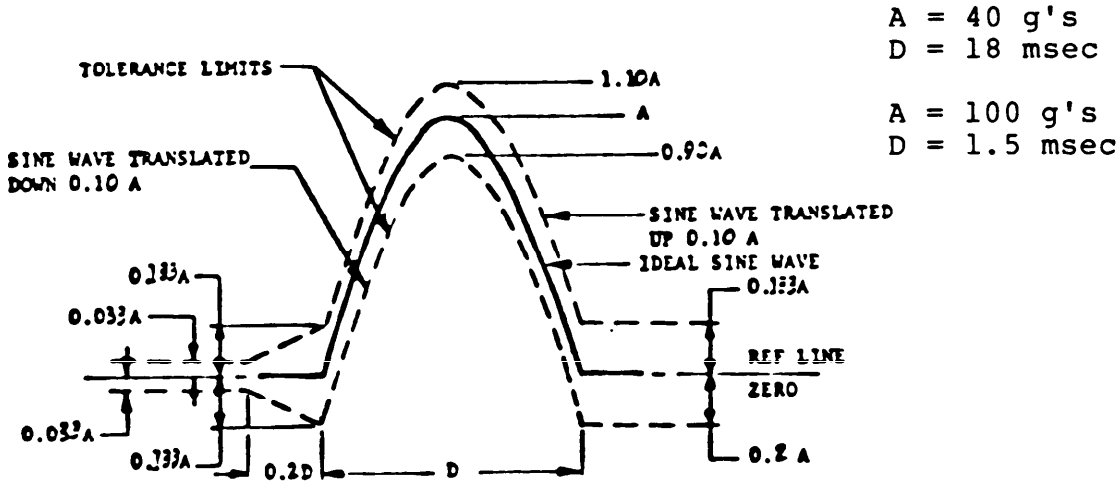


FIGURE 2. Shock Pulse.

3.4.4 Temperature cycling. The assembly shall be capable of operating as specified herein after withstanding exposure to temperature cycling and soaking between the temperatures of -25°F and 140°F in accordance with Figure 3 (see 4.6.2.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in the specification shall become a part of the contractor's overall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

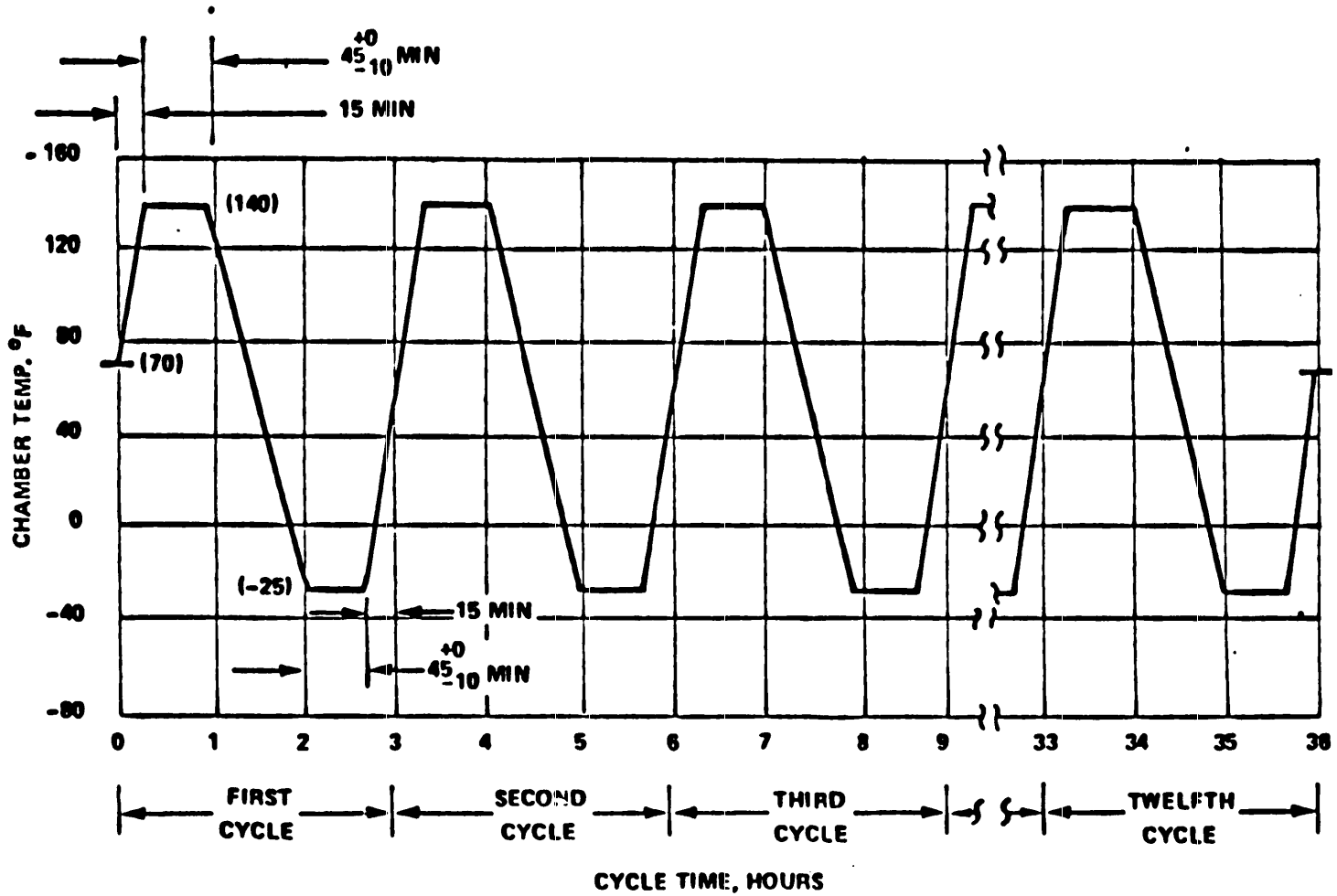


FIGURE 3. Temperature cycling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and standard commercial practice. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract (see 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.4).
b. Quality conformance inspection (see 4.5 and 6.8).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article inspection. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract (see 6.3).

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.4.2. The first article sample shall consist of two assemblies.

4.4.2 Inspections to be performed. Two assemblies shall be subjected to all the inspections specified in Tables IV, V and VI. Tests 301 and 302 of Table VI shall be performed first.

TABLE IV. Performance tests.

Item	Characteristics	Requirements	Test Procedures
101	Lamp diodes forward biased	3.3.1	4.6.1.1
102	Lamp diodes reversed biased	3.3.2	4.6.1.2
103	Scaling and bridge resistors	3.3.3	4.6.1.3
104	Laser rangefinder select logic	3.3.4	4.6.1.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE V. Environmental tests.

Item	Characteristics	Requirements	Test Procedures
201	High Temperature	3.4.1	4.6.2.1
202	Low Temperature	3.4.1	4.6.2.2
203	Vibration	3.4.2	4.6.2.3
204	Shock	3.4.3	4.6.2.4
205	Temperature cycling	3.4.4	4.6.2.5

TABLE VI. General tests.

Item	Characteristics	Requirements	Test Procedures
301	Fabrication	3.1	Applicable drawings - Visual
302	General specifications	3.1.2	4.6.3 - Visual
303	Packaging inspection	5.1	4.6.4 - Visual

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A, B and C
b. Component assembly of the next higher assembly	B and C

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.5.2 Inspection plan A.

4.5.2.1 Environmental sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in Table V shall be applied separately to the assembly.

4.5.2.2 Acceptance. When the sample assembly of the quantity produced fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted for reinspection only after all assemblies are re-examined or retested and all defective assemblies are removed or defects corrected.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative. All the tests in Table VI and only the test of item 205 in Table V shall apply. The following sampling plan (except that the visual inspection for completeness, improper assembly and evidence of poor workmanship per items 301 and 302 shall be performed 100%) shall be utilized:

100%, c=0 for lot sizes of 10 or less
n=10, c=0 for lot sizes of 11 to 150
n=32, for lot sizes greater than 150

where, n=sample size and c=0 requires zero defects during inspection for acceptance

4.5.3.2 Acceptance. Where any one assembly in the sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted as above.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in Table IV and shall be examined visually for completeness, improper assembly and evidence of poor workmanship per items 301 and 302 of Table VI. The conditions of 3.1.3 apply.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.6 Methods of Inspection.

4.6.1 Performance tests.

4.6.1.1 Lamp diodes forward biased. Apply load, power source, and interconnect per 3.3.1. Verify load lamp conditions per 3.3.1.

4.6.1.2 Lamp diodes reverse biased. Apply load, power source, and interconnect per 3.3.2. Verify load lamp condition per 3.3.2.

4.6.1.3 Scaling and bridge resistors. Verify resistance values as specified in 3.3.3.

4.6.1.4 Laser rangefinder select logic. Apply power sources and interconnect per 3.3.4. The output shall be measured and verified to comply with the requirements of 3.3.4.

4.6.2 Environmental tests.

4.6.2.1 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, Method 501 except storage temperature shall be 160°F for 48 hours. At the conclusion of the storage temperature test, stabilize the assembly at 100°F and subject it to the performance tests of Table IX. After testing, return the assembly to room temperature and subject it to the performance tests of Table IV.

4.6.2.2 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, Method 502. Maintain the temperature at -65°F for a period of 12 hours minimum. At the conclusion of the storage temperature test, stabilize the assembly at 40°F and subject it to the performance tests of Table IV. After testing, return the assembly to room temperature and subject it to the performance tests of Table IV.

4.6.2.3 Vibration. Vibration A as specified in 4.6.2.3.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.2.3.2 shall be conducted only for quality conformance inspection.

4.6.2.3.1 Vibration A. Mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, Method 514, General Procedures, and 3.4.2. The vibration amplitude shall have a tolerance of ± 10%. The tolerance of the vibration frequency shall be ± 2%, or 1/2 Hz for frequencies below 25 Hz. Vibration sweeps shall be conducted with the duration of exposure not less than 80 minutes in each of three mutually perpendicular axes. At the conclusion of this test, subject the assembly to the performance tests of Table IV.

4.6.2.3.2 Vibration B. This test shall be conducted as in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.6.2.3.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.2.4 Shock. Mount the assembly in a suitable shock fixture and subject it to the functional shock test for ground equipment specified in MIL-STD-810, Method 516, for a total of three half sine wave shock pulses in both directions along three mutually perpendicular axes (18 shocks). Peak amplitude shall be 40 ± 4 g's with a time duration of 18 ± 3 msec in accordance with Figure 2. At the conclusion of this test, subject the assembly to the performance tests of Table IV. Repeat this test with peak amplitude of 100.0 ± 10.0 g's with a time duration of 1.5 ± 0.2 msec in accordance with Figure 2.

4.6.2.5 Temperature cycling. With power and loads of Table I applied, subject the assembly to the temperature profile specified in Figure 3. At the conclusion of this test subject the assembly to the performance tests of Table IV.

4.6.3 General specifications. Verification shall be provided by examination, visually or through the use of standard measuring equipment, in accordance with 3.1.2.

4.6.4 Packaging inspection. The sampling and inspection of the preservation, packaging, packing and container marking shall be in accordance with MIL-STD-2073-1.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-2073-1. The level of protection shall be as specified in the procurement document (see 4.6.4 and 6.2d).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Lamp Diodes circuit card (A11) assembly is intended for use in the Field Test Set, 11733200, which is used in the field for test of all of the M21 Computer subsystem components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

c. Inspection provisions for first article (see 6.3).

d. Applicable levels of preservation, packaging, packing and container marking (see 5.1 and 6.7).

6.3 First article. The contracting officer should provide specific guidance to offerors for the submission of a first article sample to be tested as specified in 4.4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Supersession. Previous revisions of MIL-C-48487(AR) were for the US Army Armament, Research, Development and Engineering Center's Internal Use Only, and were not forwarded to the Naval Publications and Form Center for publication.

6.5 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAC-F, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Levels of packaging. The following definitions describe the three levels of packaging in accordance with MIL-STD-2073-1.

a. Level A - Adequate protection for indeterminate storage and worldwide distribution.

b. Level B - Adequate protection for known shipment conditions and storage conditions normally not exceeding one year.

c. Level C - Adequate protection for shipment to a known site for immediate use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

6.8 Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equivalent, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equivalent, the contractor shall describe them in a written proposal submitted to the Government for evaluation. When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity.

6.9 Subject term (key word) listing.

Computer, M21
Fire Control
Tank, M60A3

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-0010)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.